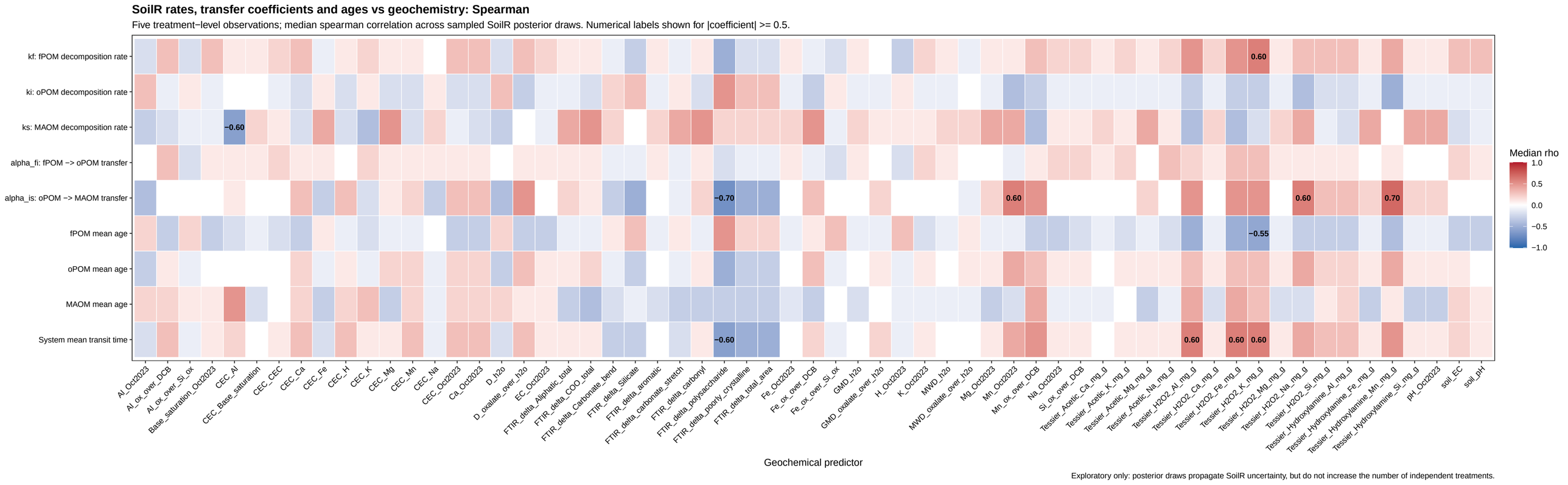
SoilR rates, transfer coefficients and ages vs geochemistry: Spearman
Five treatment−level observations; median spearman correlation across sampled SoilR posterior draws. Numerical labels shown for |coefficient| >= 0.5.
kf: fPOM decomposition rate
ki: oPOM decomposition rate
ks: MAOM decomposition rate
alpha_fi: fPOM −> oPOM transfer
alpha_is: oPOM −> MAOM transfer
fPOM mean age
oPOM mean age
MAOM mean age
System mean transit time
−0.60
0.60
−0.70
0.60
0.70
−0.55
−0.60
0.60
0.60
0.60
Al_Oct2023
Al_ox_over_DCB
Al_ox_over_Si_ox
Base_saturation_Oct2023
CEC_Al
CEC_Base_saturation
CEC_CEC
CEC_Ca
CEC_Fe
CEC_H
CEC_K
CEC_Mg
CEC_Mn
CEC_Na
CEC_Oct2023
Ca_Oct2023
D_h2o
D_oxalate_over_h2o
EC_Oct2023
FTIR_delta_Aliphatic_total
FTIR_delta_COO_total
FTIR_delta_Carbonate_bend
FTIR_delta_Silicate
FTIR_delta_aromatic
FTIR_delta_carbonate_stretch
FTIR_delta_carbonyl
FTIR_delta_polysaccharide
FTIR_delta_poorly_crystalline
FTIR_delta_total_area
Fe_Oct2023
Fe_ox_over_DCB
Fe_ox_over_Si_ox
GMD_h2o
GMD_oxalate_over_h2o
H_Oct2023
K_Oct2023
MWD_h2o
MWD_oxalate_over_h2o
Mg_Oct2023
Mn_Oct2023
Mn_ox_over_DCB
Na_Oct2023
Si_ox_over_DCB
Tessier_Acetic_Ca_mg_g
Tessier_Acetic_K_mg_g
Tessier_Acetic_Mg_mg_g
Tessier_Acetic_Na_mg_g
Tessier_H2O2_Al_mg_g
Tessier_H2O2_Ca_mg_g
Tessier_H2O2_Fe_mg_g
Tessier_H2O2_K_mg_g
Tessier_H2O2_Mg_mg_g
Tessier_H2O2_Na_mg_g
Tessier_H2O2_Si_mg_g
Tessier_Hydroxylamine_Al_mg_g
Tessier_Hydroxylamine_Fe_mg_g
Tessier_Hydroxylamine_Mn_mg_g
Tessier_Hydroxylamine_Si_mg_g
pH_Oct2023
soil_EC
soil_pH
Geochemical predictor
Median rho
1.0
0.5
0.0
−0.5
−1.0
Exploratory only: posterior draws propagate SoilR uncertainty, but do not increase the number of independent treatments.